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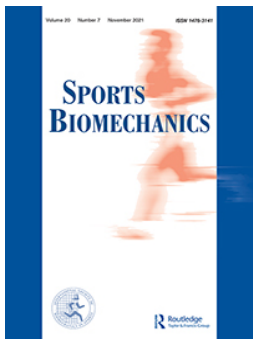
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Validity and reliability of linear position transducers and linear velocity transducers: a systematic review

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ABSTRACT

This systematic review aimed to summarise and analyse the evidence on the reliability and validity of linear transducers (LTs) in exercises of different nature and different modes of execution. This systematic review was carried out under PRISMA guidelines, and was carried out using three databases (PubMed, Web of Sciences, and Scopus). Of the 351 initially found, 21 were included in the qualitative synthesis. The results reflected that linear position transducers (LPTs) were valid and reliable in monitoring movement velocity in non-plyometric exercises. However, precision and reliability were lower in execution protocols without isometric phase and in the execution of exercises in multiple planes of movement, with greater measurement errors at higher sampling frequencies. On the other hand, linear velocity transducers (LVTs) proved to be valid and reliable in measuring velocity during plyometric and non-plyometric exercises performed on the Smith machine, with less variation in measurement in the latter. Finally, the use of peak values is recommended, since they are less dependent on the technological errors of LTs. Therefore, the performance of non-plyometric exercises, carried out in the Smith machine and with an isometric phase in the execution of the movement, will help to minimise the technological error of the LTs.

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Accuracy; agreement; encoders; repeatability; reproducibility

Introduction

Resistance training is considered a fundamental component in the physical preparation of the athlete, where velocity-based training (VBT) has been postulated as one of the main monitoring methods (González-Badillo & Sánchez-Medina, 2010; Izquierdo et al., 2006; Pereira & Gomes, 2003; Sánchez-Medina & González-Badillo, 2011). In fact, since strong correlation has been shown between execution velocity and 1RM (García-Ramos, Pestaña-Melero et al., 2018; Muñoz-López et al., 2017; Pérez-Castilla et al., 2020), the VBT monitoring allows the creation of individualised velocity profile (Alcazar et al., 2017; Banyard et al., 2018), as well as the degree of fatigue (García-Ramos, Torrejón et al., 2018; Haff & Nimphius, 2012; Sánchez-Medina & González-Badillo, 2011). Hence, VBT facilitates the prescription of personalised velocity training zones and the knowledge of the athlete's physiological state (Conceição et al., 2016; González-Badillo et al., 2015;

González-Badillo & Sánchez-Medina, 2010) in a robust, non-invasive, and highly sensitive way (González-Badillo & Sánchez-Medina, 2010; Martínez-Cava et al., 2019; Morán-Navarro et al., 2019; Sánchez-Medina & González-Badillo, 2011; Sánchez-Medina et al., 2017).

However, to offer a safe and effective VBT prescription, those technologies should be valid and reliable (Martínez-Cava et al., 2020), making difficult the decision of what technology should be used among a wide range of devices. Between the different commercially available VBT technologies, the three-dimensional (3D) motion capture system and force platforms are usually considered as the most accurate systems (Giroux et al., 2015; Lorenzetti et al., 2017; Rahmani et al., 2001; Sato et al., 2015). However, due to the incremental grown of manufacturers and branches, together with the non-practical viability or cost-effective issues (Cronin et al., 2004; Walsh et al., 2006), other type of technologies have been validated (Weakley et al., 2021). For example, linear position transducers (LPT) are postulated as a practical and affordable alternative for the quantification of velocity, power, and force variables in a practical work environment (Balsalobre-Fernández et al., 2016; Banyard et al., 2017; García-Ramos et al., 2017; Garnacho-Castaño et al., 2015), and they have become the most commonly used LTs in the scientific and practical field (Balsalobre-Fernández et al., 2016; Banyard et al., 2017; García-Ramos et al., 2017). The LPT consists of an isoinertial dynamometer with a cable that is usually connected to the bar, and directly measures the vertical displacement of the cable, deriving the displacement time using the inverse dynamics approach to obtain the velocity (Harris et al., 2010; Pérez-Castilla et al., 2019). Among other alternatives, linear velocity transducers (LVT) have recently been developed that provide velocity measurements directly, by registering electrical signals that are proportional to the velocity of cable extension (McMaster et al., 2014; Sánchez-Medina & González-Badillo, 2011). The validation of these devices is necessary in a wide variety of exercises, with different execution models, as well as different load ranges (Mitter et al., 2019).

In this scenario, both LPTs and LVTs have different limitations inherent to their technology, which can alter the resulting velocity data depending on the exercises evaluated and the way they are performed (Król & Gołaś, 2017; Swinton et al., 2012, 2011). To date, despite the wide spectrum of linear transducers' (LTs) validation articles (Courel-Ibáñez et al., 2019; Crewther et al., 2011; Fernandes et al., 2018; Gonzalez et al., 2019; Mitter et al., 2019; Pérez-Castilla et al., 2019), to the author's knowledge, there is only one systematic review in this regard (Weakley et al., 2021). In addition, this study (Weakley et al., 2021) did not differentiate between LPTs and LVTs. This is a relevant aspect since the technologies of each type of device at the time of sampling and data treatment can mark differences in the final values (Harris et al., 2010; McMaster et al., 2014; Pérez-Castilla et al., 2020; Sánchez-Medina & González-Badillo, 2011). Furthermore, the authors only took into account whether the LTs were compared with a gold standard and the plane in which the exercise was performed (2D or 3D), without investigating factors such as the nature (plyometric or non-plyometric) of the evaluated exercises nor the mode of execution of the movement (contraction regimes). These variables have been shown to alter the kinetic and kinematic parameters (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012, 2011). Therefore, the objective of this systematic review is to summarise and analyse the validity and reliability of LPT and LVT, in exercises of a diverse nature and with different forms of execution.

Method

Experimental approach to the problem

A systematic review was performed in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta Analyses) guidelines (Moher et al., 2009) and guidelines for performing systematic reviews in sports science (Rico-González et al., 2021).

Search strategy

A systematic search of three databases (PubMed, Web of Sciences and Scopus) was performed to identify articles published prior to 15 February 2021. Keywords and synonyms were entered in various combinations in the title, abstract or keywords: (validity OR reliability OR precision OR agreement) AND (transducer OR encoder) AND (sport OR exercise). Additionally, the reference lists of the studies retrieved were manually searched to identify potentially eligible studies not captured by the electronic searches.

Screening strategy and study selection

When the referred authors had completed the search (A.M.V., M.R.G. and J.P.O.), they compared the results among themselves to ensure that the same articles were identified. Then, one of the authors (A. M. V.) downloaded and put the main data from the articles (title, authors, date, and database) onto an Excel spreadsheet (Microsoft Excel, Microsoft, Redmond, USA). Then, two authors (A.M.V. and J.P.O) removed duplicates. The remaining articles were screened and checked by two authors independently (A.M.V. and M.R.G.) against the inclusion and exclusion criteria (Table 1). Moreover, relevant articles not previously identified were also screened in an identical manner and further studies that complied with the inclusion-exclusion criteria were included and labelled as ‘included from external sources’.

Data analysis

The information collected from the selected studies addressed the following aspects: (i) validity and (ii) reliability of the LPT and LVT in the variables of velocity, power and force (mean velocity, peak velocity, mean power, peak power, mean force, peak force). In addition, the following data items were extracted: (i) characteristics of the participants (number of subjects, age, sex, experience); (ii) characteristics of the linear transducers and comparator (for the case of validity studies); (iii) characteristics of the experimental protocol (selected exercises, execution movement, training loads).

Quality of studies

Two authors (AMV and MRG) performed the methodological assessment of the studies eligible for inclusion using an adapted version of the STROBE assessment criteria, as was used in (O’Reilly et al. (2018)). Hence, each article was evaluated using 10 specific criteria. If any disagreement appears, it was discussed and solved by consensus decision. The

Table 1. Inclusion/exclusion criteria.

	Inclusion	Exclusion
Population	Tests were conducted in healthy athletes or recreationally healthy active adults, with experience in force (e.g., resistance training exercises) and/or power (e.g., jumping, lifting) movements.	The tests were not conducted in athletes (e.g., pregnancy, elderly) or in health active adults (i.e., injury) and/or without experience in resistance or power movements.
Intervention	Concerned with the validity and/or reliability of commercially available resistance training linear velocity and/or position transducers that monitor force, velocity, and power outputs during resistance training.	Not concerned with validity or reliability of linear velocity and/or position transducers, or not assessed in resistance training.
Comparator	In the case of validity, the transducers were compared to recognised gold-standard (motion capture systems, force plate, linear position transducer and/or linear velocity transducer)	For validity, the transducers were not compared to recognised gold standard methods or compared to other transducers.
Outcomes	Studies that describe the velocity of execution of the evaluated movement. In the case of validity, one of the following measures were included: (i) intraclass correlation coefficient; (ii) correlation coefficient; (iii) standard error of the estimate; and (iv) minimal products regression In the case of reliability, one of the following measures were included: (i) intraclass correlation coefficient; (ii) typical error of measurement; (iii) coefficient of variation; and (iv) standard error of measurement.	Studies that do not describe the velocity of execution of the movement evaluated. For validity, outcomes presented are not intraclass correlation coefficient, correlation coefficient, standard error of estimate, or minimal products regression. For reliability, outcomes presented are not intraclass correlation coefficient, typical error of measurement, coefficient of variation, or standard error of measurement.
Study	Original research and full-text studies	Other article types than original (e.g., reviews, letters to editors, trial registrations, proposals for protocols, editorials, book chapters and conference abstracts).

study rating was qualitatively interpreted following O'Reilly et al. (2018): from 0 to 7 scores, the study was considered as risk of bias (low quality); while, if the study was rated from 8 to 10 points, it was considered as a low risk of bias (high quality).

Results

Identification and selection of studies

A total of 347 (i.e., PubMed: 155; WoK: 83; Scopus: 109) original articles were initially retrieved from the mentioned databases, of which 128 were duplicates. Thus, a total of 219 original articles were found. After this, a total of 171 articles checked by title and abstract were excluded. The remaining 48 articles were checked in full text, leading to the exclusion of 30 articles according to criterion nº 2, and one according to criterion nº 3. Additionally, 4 articles were included from additional sources. A total of 21 articles met all the inclusion criteria and were finally included in the qualitative synthesis. All the steps followed for the selection of the articles is available in [Figure 1](#).

Methodological quality

The overall methodological quality of the cross-sectional studies can be found in [Table 2](#).

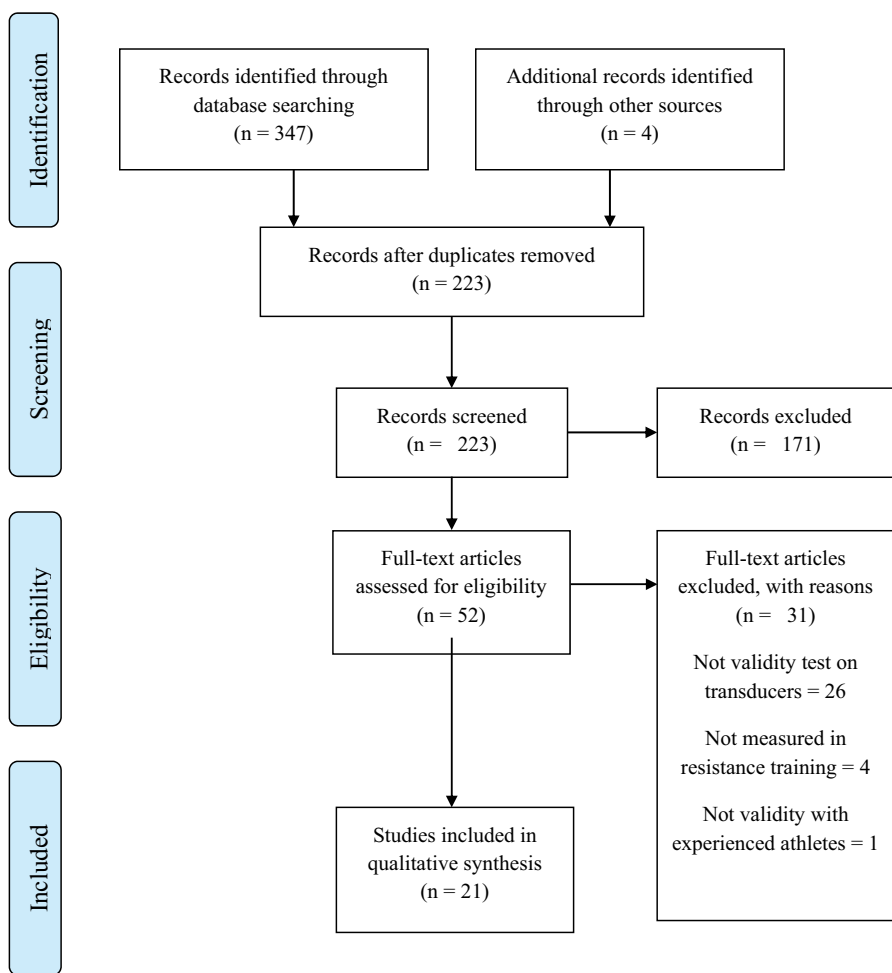


Figure 1. PRISMA flow diagram.

Study characteristics

The characteristics of the study intervention protocols are available in Table 3.

Of the 21 articles included, eight investigated validity and reliability (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; García-Ramos et al., 2016; Garnacho-Castaño et al., 2015; Lorenzetti et al., 2017; Pérez-Castilla et al., 2019; Thompson et al., 2020), while six tested only validity (Crewther et al., 2011; Fernandes et al., 2018; Gonzalez et al., 2019; , 2018; Mitter et al., 2019; Pérez-Castilla et al., 2017) and seven only tested reliability (Ferro et al., 2019; Grgic et al., 2020; Hansen et al., 2011a, 2011b; Jennings et al., 2005; Martínez-Cava et al., 2020; Orange et al., 2020).

Regarding the devices evaluated, 18 studies included LPTs such as GymAware (Askow et al., 2018; Crewther et al., 2011; Dorrell et al., 2019; Grgic et al., 2020; Lorenzetti et al., 2017; Mitter et al., 2019; Orange et al., 2020; Thompson et al., 2020), FitroDyne (fitronic) (Fernandes et al., 2018; Jennings et al., 2005; Mitter et al., 2019), Speed4lift (Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019), Chronojump (Courel-Ibáñez et al., 2019;

Table 2. Methodological assessment of the included studies.

Study (n = 21)	1	2	3	4	5	6	7	8	9	10	Total score	Quality
Askow et al. (2018)	1	1	1	1	1	1	1	1	1	1	10	High
Courel-Ibáñez et al. (2019)	1	0	0	1	1	1	1	1	1	0	7	Low
Crewther et al. (2011)	1	1	1	1	1	1	1	1	0	1	9	High
Dorrell et al. (2019)	1	0	1	1	1	1	1	0	1	0	7	Low
Fernandes et al. (2018)	1	0	1	1	1	1	1	0	1	1	8	High
Ferro et al. (2019)	1	1	0	1	1	1	1	1	1	1	9	High
García-Ramos et al. (2016)	1	1	0	1	1	1	1	0	1	1	8	High
Garnacho-Castaño et al. (2015)	1	1	1	1	1	1	1	1	0	0	8	High
Gonzalez et al. (2019)	1	1	0	1	1	1	1	1	1	1	9	High
Grgic et al. (2020)	1	1	1	1	1	1	1	1	0	1	9	High
Hansen et al. (2011a)	1	0	0	1	1	1	1	0	1	0	6	Low
Hansen et al. (2011b)	1	0	0	1	1	1	1	1	1	0	7	Low
Jennings et al. (2005)	1	0	1	1	1	1	1	0	0	1	7	Low
Lorenzetti et al. (2017)	1	0	0	1	1	1	1	1	1	1	8	High
Martínez-Cava et al. (2020)	1	0	1	1	1	1	1	1	1	0	8	High
McGrath, 2018	1	0	1	1	1	1	1	0	1	0	7	Low
Mitter et al. (2019)	1	0	1	1	1	1	1	1	1	1	9	High
Orange et al. (2020)	1	0	1	1	1	1	1	0	1	1	8	High
Pérez-Castilla et al. (2017)	1	0	0	1	1	1	1	0	1	1	7	Low
Pérez-Castilla et al. (2019)	1	1	1	1	1	1	1	1	1	1	10	High
Thompson et al. (2020)	1	1	1	1	1	1	1	1	1	1	10	High
Total item scores	21	9	13	21	21	21	21	13	17	14		

Provide in the abstract an informative and balanced summary of what was done and what was found (item 1); state specific objectives, including any prespecified hypotheses (item 2); Give the eligibility criteria, and the sources and methods of selection of participants (item 3); for each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group (item 4); explain how quantitative variable were handled in the analyses. If applicable, describe which groupings were chosen and why (item 5); give characteristics of study participants (item 6); summarise key results with reference to study objectives (item 7); discuss limitations of the study, considering sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias (item 8); give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence (item 9); give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based (item 10).

Pérez-Castilla et al., 2019), Tendo FitroDyne (, 2018), HX VPA-200 (Hansen et al., 2011a, 2011b), Tendo Weightlifting Analyser System (Garnacho-Castaño et al., 2015; Lorenzetti et al., 2017) and Open Barbell (Gonzalez et al., 2019). Additionally, five studies analysed LVTs such as T-Force (García-Ramos et al., 2016; Lorenzetti et al., 2017; Pérez-Castilla et al., 2017, 2019) and SmartCoach (Ferro et al., 2019).

The validity and reliability of kinematic variables such as velocity (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Ferro et al., 2019; García-Ramos et al., 2016; Garnacho-Castaño et al., 2015; Gonzalez et al., 2019; Grgic et al., 2020; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; , 2018; Mitter et al., 2019; Orange et al., 2020; Pérez-Castilla et al., 2017, 2019; Thompson et al., 2020), power (Askow et al., 2018; Crewther et al., 2011; García-Ramos et al., 2016; Garnacho-Castaño et al., 2015; Grgic et al., 2020; Jennings et al., 2005; Orange et al., 2020; Pérez-Castilla et al., 2017) and force (Askow et al., 2018; Crewther et al., 2011; Dorrell et al., 2019; García-Ramos et al., 2016; Hansen et al., 2011b; Pérez-Castilla et al., 2017) were investigated using both peak and mean values. In a marginal way, some studies also evaluated acceleration variables (Courel-Ibáñez et al., 2019; Hansen et al., 2011a, 2011b; Lorenzetti et al., 2017; Martínez-Cava et al., 2020). These variables were evaluated by comparing the encoder with a criterion measurement device (in the case of validity). Thus, motion capture systems (Askow et al., 2018; Dorrell et al., 2019; Lorenzetti et al., 2017; , 2018;

Table 3. Study characteristics.

Study	Outcome tested	Tested validity	Tested reliability	Device			Comparator characteristics	Population characteristics	Measured exercises	Experimental protocol	Movement execution
				Device	characteristics (sample rate, version)	characteristics (sample rate, version)					
Linear Position Transducers											
Kinetic Performance Technology											
Askow et al. (2018)	Peak velocity	Yes	Yes	GymAware	Variable sample rate, 50 Hz	Qualysis Motion Capture System (120 Hz sample rate) & AMTI Force Plate (1200 Hz sample rate)	9 males (Age: 24.3 ± 5.6; 82.5 ± 9.6 kg; 174 ± 10 cm) Exp: > 2 years	Back squat (FW)	2 attempts at 75 and 90% 1 RM until fatigue	Vmáx concentric	
	Mean velocity										
	Peak force										
	Mean force										
	Peak power										
	Mean power										
Crewther et al. (2011)	Peak force	Yes	No	GymAware	Variable sample rate, 50 Hz	Force plate (Kistler; 1000 Hz sample rate)	12 males (Age: 28.8 ± 6.8; 86.8 ± 9.2 kg; 181 ± 8 cm) Exp: Yes	Loaded jump squat (FW)	2 repetitions at 20, 40, 60 and 80 kg	Vmáx concentric + ecc controlled	
	Peak power				downsample rate						
Dorrell et al. (2019)	Peak velocity	Yes	Yes	GymAware	Variable sample rate, 50 Hz	Rapture-E 3D cameras (150 Hz sample rate) and Force Plate (Kistler; 1500 Hz sample rate)	13 males (Age: 26.5 ± 4.8; 81.9 ± 12.1 kg; 174 ± 10 cm) Exp: > 2 years	Back squat (FW)	3 repetitions at 80% 1RM for each exercise	Vmáx concentric + eccentric controlled	
	Mean velocity			PowerTool	downsample rate			Bench press (FW)			
	Peak force							Deadlift (FW)			
	Mean force										
Grgic et al. (2020)	Peak velocity	No	Yes	GymAware	Variable sample rate, 50 Hz	-	10 males and 6 females (Average age: 26.5) Exp: > 1 year	Squat (-)	3 repetitions at 30, 45, 60 and 75% 1RM, and 1 repetition at 90% 1RM, for each exercise	Vmáx concentric + eccentric controlled	
	Mean velocity				downsample rate			Deadlift			
	Peak power										
	Mean power										

(Continued)

Table 3. (Continued).

Study	Outcome tested	Tested validity	Tested reliability	Device		Comparator characteristics	Population characteristics	Measured exercises	Experimental protocol	Movement execution
				Device	characteristics (sample rate, version)					
Lorenzetti et al. (2017)	Peak velocity Mean velocity Time to peak velocity	Yes	Yes	GymAware	Variable sample rate, 50 Hz downsample rate (v 1.1.2)	Vicon 3D (100 Hz sample rate)	9 males (Age: 30.9 ± 5.9; 92.0 ± 8.7 kg; 182 ± 6 cm) Exp: 9.7 ± 5.5 years	Back squat Jump squat	2 x 5 repetitions at 70% 1RM for back squat, and 2 x 5 repetitions at 25 kg for jump squat	V _{max} concentric
Mitter et al. (2019)	Peak velocity Mean velocity	Yes	No	GymAware	Variable sample rate, 50 Hz downsample rate (v. 4)	Vicon 3D (200 Hz sample rate)	18 males and 6 females (Age: 25.1 ± 5.1; 90.9 ± 20.0 kg) Exp: > 2 years	Back squat (FW) Bench press (FW) Deadlift (FW)	1–3 repetitions at 30, 50, 60, 70, 80 y 90% 1RM	V _{max} concentric + ecc controlled + 0.5 seconds isometric
Orange et al. (2020)	Peak velocity Mean velocity Peak power Mean power	No	Yes	GymAware	Variable sample rate, 50 Hz downsample rate (v 2.1.1)	-	29 males (Age: 17.6 ± 1.0; 87.3 ± 20.8 kg; 173 ± 18 cm) Exp: 3.1 ± 1.3 years	Back squat (FW) Bench press (FW)	3 repetitions at 20, 40, 60 and 80% 1RM, and 2 repetitions at 90% 1RM	V _{max} concentric
Thompson et al. (2020)	Peak velocity Mean velocity	Yes	Yes	GymAware	Variable sample rate, 50 Hz downsample rate	12 Camera Raptor 3D Motion Capture (250 Hz sample rate)	10 males (Age: 25.0 ± 5.6; 73.6 ± 13.9 kg; 169 ± 6 cm) Exp: > 1 year (élite)	Back squat (FW) Power clean	3 repetitions at 60% 1RM, 2 repetitions at 70 and 80% 1RM, and 1 repetition at 90% 1RM	V _{max} concentric
<i>Boscosystem</i>										
Courel-Ibáñez et al. (2019)	Peak velocity Mean velocity Mean propulsive velocity	Yes	Yes	Chronojump	500 Hz (v. 1.7.1) v. 2.60; 1000 Hz sample rate)	T-Force (LVT; v. 2.60; 1000 Hz sample rate)	17 males (Age: 26.0 ± 3.6; 81.5 ± 6.8 kg; 178 ± 8 cm) Exp: 11 ± 4 years	Back squat (SM) Bench press (SM) Prone bench pull (SM)	2 repetitions at 20, 30, 40, 50, 60, 70 and 80 kg for each exercise	V _{max} concentric + 0.5–0.7 seconds eccentric + 1.5 seconds isometric

(Continued)

Table 3. (Continued).

Study	Outcome tested		Tested validity	Tested reliability	Device		Comparator characteristics	Population characteristics	Measured exercises	Experimental protocol	Movement execution
	Mean velocity				Device	characteristics (sample rate, version)					
Pérez-Castilla et al. (2019)	Mean velocity		Yes	Yes	Chronojump	1000 Hz (v.1.6.2)	Trio OptiTrack (Optical motion system; 120 Hz sample rate)	14 males (Age: 22.9 ± 1.6; 76.9 ± 7.8 kg; 176 ± 6 cm) Exp: Yes	Bench press (SM)	3 repetitions at 45, 55, 65, 75 and 85% 1RM	Vmáx concentric
Fitronic											
Fernandes et al. (2018)	Peak velocity Mean velocity		Yes	No	FitroDyne	100 Hz	GymAware (LPT; variable sample rate reducing to 50 points per second)	15 males (Age: 31.4 ± 12.2; 84.6 ± 14.8 kg; 180 ± 10 cm) Exp: > 2 years	Bench press (SM) Back squat (SM) Bent-over row (SM)	3 repetitions at 20, 30, 40, 50, 60, 70 and 80% 1RM for each exercise	Vmáx concentric
Jennings et al. (2005)	Peak power		No	Yes	FitroDyne	100 Hz	-	30 males (Age: 24.8 ± 5.9; 76.5 ± 11.0 kg; 178 ± 6 cm) Exp: > 3 months	Biceps Curl Scott Squat jump (FW)	3 repetitions at 0, 20, 40, 60, 80 and 90% 1RM for each exercise	Vmáx concentric + 1 second eccentric + isometric pause
Mitter et al. (2019)	Peak velocity Mean velocity		Yes	No	FitroDyne	100 Hz (v 2.1)	Vicon 3D (200 Hz sample rate)	18 males and 6 females (Age: 25.1 ± 5.1; 90.9 ± 20.0 kg) Exp: > 2 years	Back squat (FW) Bench press (FW) Deadlift (FW)	1–3 repetitions at 30, 50, 60, 70, 80 y 90% 1RM	Vmáx concentric + ecc controlled + 0.5 seconds isometric
Tendo Sports Machines											
Garnacho-Castaño et al. (2015)	Peak velocity Mean velocity Peak power Mean power		Yes	Yes	Tendo Weightlifting Analyser System	Frequency determined by speed of rotation of the disk (v.3.0.4)	T-Force (LVT; 1000 Hz sample rate)	71 males (Age: 21.15 ± 2.75; 74.1 ± 9.6 kg; 174 ± 6 cm) Exp: 3.2 ± 1.7 years	Back squat (SM) Bench press (SM)	4 repetitions at 40 kg, 3 repetitions at 50 kg, 2 repetitions at 60 kg, and a set until muscle failure at 85% 1RM	Vmáx concentric + 1 second eccentric
Lorenzetti et al. (2017)	Peak velocity Mean velocity Time to peak velocity		Yes	Yes	Tendo	Sample rate determined by speed of rotation of the disc (v 4.1.0)	Vicon 3D (100 Hz sample rate)	9 males (Age: 30.9 ± 5.9; 92.0 ± 8.7 kg; 182 ± 6 cm) Exp: 9.7 ± 5.5 years	Back squat Jump squat	2 x 5 repetitions at 70% 1RM for back squat, and 2 x 5 repetitions at 25 kg for jump squat	Vmáx concentric

(Continued)

Table 3. (Continued).

Study	Device characteristics				Tested validity	Tested reliability	Device	Comparator characteristics	Population characteristics	Measured exercises	Experimental protocol	Movement execution
	Outcome tested	Frequency (sample rate, version)	3D Motion	Camera Eagle								
, 2018	Mean velocity	Frequency determined by speed of rotation of the disk	3D Motion	Camera Eagle	Yes	No	Fitrodyn	200 Hz sample rate)	10 males (Age: 23.4 ± 6.8; 84.2 ± 8.8 kg; 178 ± 6 cm) Exp: > 1 year	Bench press (FW)	6 repetitions at 40 and 80% 1RM	Vmáx concentric + eccentric controlled + isometric pause
Squats & Science												
Gonzalez et al. (2019)	Peak velocity	Velocity calculation every 2.8 millimetres	Tendo (LPT; frequency determined by speed of rotation of the disk)	Open Barbell System	Yes	No			11 males (Age: 19.4 ± 1.0; 87.2 ± 7.4 kg; 182.4 ± 6.5 cm) Exp: 5.3 ± 2.2 years	Back squat (FW)	3 repetitions at 30, 50, 70% 1RM and 2 repetitions at 90% 1RM	Vmáx concentric
	Mean velocity									Front squat (FW)		
Unimeasure												
Hansen et al. (2011a)	Time to peak velocity	500 Hz	Force Plate (Accupower)	HX VPA-200	No	Yes			25 males (Age: 24.4 ± 4.9; 98.6 ± 12 kg; 180 ± 10 cm) Exp: Yes (elite)	Loaded squat jump	3 repetitions at 40 kg	Pause between jumps
	Time to peak power											
Hansen et al. (2011b)	Rate of force development											
	Peak force	500 Hz	Force Plate (Accupower)	HX-VPA-200	No	Yes			25 males (Age: 23.6 ± 4.8; 180 ± 10 cm) Exp: Yes	Loaded squat jump	3 repetitions at 40 kg	Pause between jumps
	Time to peak force											
	Rate of force development											
Speed4Lifts												
Martínez-Cava et al. (2020)	Peak velocity	100 Hz (v. 2.0)	T-Force (LVT; 1000 Hz sample rate; v.1.41)	Speed4Lifts	No	Yes			15 males (Age: 27.0 ± 3.8; 78.8 ± 7.6 kg; 178 ± 6 cm) Exp: Yes	Back squat (SM)	1 repetition at 25, 35, 45, 55, 65, 75, 85, 95 kg for each exercise	Vmáx concentric + 0.5–0.7 seconds eccentric
	Mean velocity									Bench press (SM)		
	Mean propulsive velocity											

(Continued)

Table 3. (Continued).

Study	Outcome tested	Tested validity	Tested reliability	Device	Device characteristics (sample rate, version)	Comparator characteristics	Population characteristics	Measured exercises	Experimental protocol	Movement execution
Pérez-Castilla et al. (2017)	Peak velocity Peak force Peak power	Yes	No	T-Force	1000 Hz	Force Plate (Dinascan; 1000 Hz sample rate)	23 males (Age: 23.1 ± 3.2; 74.7 ± 7.3 kg; 177 ± 7 cm) Exp: Yes	Restricted squat jump (SM) No-restricted squat jump (SM)	2 repetitions at 17, 30, 45, 60 and 75 kg	Vmáx concentric + 1.5 seconds eccentric + 1.5 seconds isometric
Pérez-Castilla et al. (2019)	Mean velocity	Yes	Yes	T-Force	1000 Hz (v.2.28)	Trio OptiTrack (Optical motion system; 120 Hz sample rate)	14 males (Age: 22.9 ± 1.6; 76.9 ± 7.8 kg; 176 ± 6 cm) Exp: Yes	Bench press (SM)	3 repetitions at 45, 55, 65, 75 and 85% 1RM	Vmáx concentric

3D: Three dimensions; cm: centimetres; Exp: Experience; FW: Free Weight; Hz: hertz; kg: kilograms; LPT: Linear Position Transducer; LVT: Linear Velocity Transducer; RM: Repetition Maximum; s: seconds; SM: Smith machine; v.: Version; Vmáx: Maximum Velocity

Mitter et al., 2019; Pérez-Castilla et al., 2019; Thompson et al., 2020), force plates (Crewther et al., 2011; Dorrell et al., 2019; García-Ramos et al., 2016; Hansen et al., 2011a, 2011b; Pérez-Castilla et al., 2017), LPTs (Fernandes et al., 2018; Gonzalez et al., 2019) and LVTs (Courel-Ibáñez et al., 2019; Garnacho-Castaño et al., 2015; Martínez-Cava et al., 2020) were used as comparison methods. For the analysis of the previously mentioned variables, exercises such as back squat (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Gonzalez et al., 2019; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Mitter et al., 2019; Orange et al., 2020; Thompson et al., 2020), jump squat (Crewther et al., 2011; Ferro et al., 2019; García-Ramos et al., 2016; Hansen et al., 2011a, 2011b; Jennings et al., 2005; Lorenzetti et al., 2017; Pérez-Castilla et al., 2017), front squat (Gonzalez et al., 2019), bench press (Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Martínez-Cava et al., 2020; , 2018; Mitter et al., 2019; Orange et al., 2020; Pérez-Castilla et al., 2019), deadlift (Dorrell et al., 2019; Grgic et al., 2020; Mitter et al., 2019), prone bench pull (Courel-Ibáñez et al., 2019), bent-over row (Fernandes et al., 2018), power clean (Thompson et al., 2020) and biceps curl (Jennings et al., 2005) were selected.

The validity and reliability results of the LPT and LVT devices, as well as the lessons learned and concluding remarks from the studies are available in [Tables 4 and 5](#), respectively.

Discussion and implications

To achieve an accurate, adequate and effective quantification of training loads, it is necessary that the use of technology is valid and reliable. Otherwise, there is a risk of collecting confusing data and misinterpretations that can affect the quality of training, and therefore the athlete's health and performance. Therefore, the purpose of this systematic review was to analyse the reliability and validity of LTs, in exercises of a diverse nature and with different forms of execution. In this regard, it was found that LPTs were valid and reliable in evaluating movement velocity in non-plyometric exercises. However, the LPTs were sensitive to the protocols of execution without isometric pause prior to concentric muscle action and to the execution of exercises in multiple planes of movement. As a result, measurement errors were maximised the higher the device sampling frequency. On the other hand, LVTs proved to be valid and reliable in the measurement of velocity during plyometric and non-plyometric exercises performed on the Smith machine, with less measurement variations in the latter.

However, given the differences in the steps necessary to calculate the velocity and variables derived from it (acceleration, force, power) (Cormie et al., 2007; Harris et al., 2010; McMaster et al., 2014; Sánchez-Medina & González-Badillo, 2011), the validation of LTs may be classified in two ways: (i) linear position transducers and (ii) linear velocity transducers.

Validity

Usually, the validity values are expressed as ICC (poor: <0.50, moderate: 0.50–0.75), good: 0.75–0.90, and excellent: >0.90 (Koo & Li, 2016)) and/or Pearson's correlation coefficient (trivial: 0.10, small: 0.10–0.29, moderate: 0.30–0.49, high: 0.50–0.69, very

Table 4. Validity of the linear transducers for the measurement of the velocity of movement of the bar, power and force.

Ref.	Device	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
				Linear Position Transducers					
Kinetic Performance Technology									
Askow et al. (2018)	GymAware	Back squat (FW)	SEE	0.05 m/s	0.04 m/s	PP: 80.3 W MP: 77.4 W	PF: 101.5 N MF: 45.8 N		Valid device, except for high peak velocities. Not interchangeable, since it overestimates mean variables, while it underestimates peak variables. Moderate to strong validity for the measurement of peak force and peak power. Relatively large random errors, with a load of 20 kg. Highly valid device. Caution in the deadlift (peak velocity more robust than mean velocity).
		Jump squat (FW)	r			PP: 0.62–0.82	PF: 0.59–0.87		
Dorrell et al. (2019)	GymAware	Back squat (FW)	R ²	0.99	0.99				Questions about whether a device is “valid” clearly depend on the research question, respectively, on the magnitude of the values to be quantified. High precision from GymAware for the measurement of mean and peak velocities.
		Bench press (FW)	R ²	0.99	0.93				
		Deadlift (FW)	R ²	0.99	0.54				
		Bench press (SM)	SEE	0.05 m/s	0.04 m/s			MPV: 0.997 0.06 m/s MPV: 0.994 0.03 m/s	
Lorenzetti et al. (2017)	GymAware	Prone bench pull (SM)	SEE	0.05 m/s	0.04 m/s			TTPV: 0.042 m/s TTPV: 0.046 m/s	
		Back squat	SEE	0.163 m/s	0.064 m/s				
		Jump squat	SEE	0.304 m/s	0.160 m/s				
Mitter et al. (2019)	GymAware	Back squat (FW)	SEE	0.019 m/s	0.024 m/s				
		Bench press (FW)	SEE	0.014 m/s	0.030 m/s				
		Deadlift (FW)	SEE	0.017 m/s	0.029 m/s				

(Continued)

Table 4. (Continued).

Ref.	Device	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
Thompson et al. (2020)	GymAware	Back squat (FW)	R ²	0.96–0.99	0.95–0.99				Device turned out to be valid. R ² was poorer in mean velocity with heavier loads (90% 1RM)
		Power clean	R ²	0.86–0.95	0.42–0.95				
Boscosystem									
Courel-Ibáñez et al. (2019)	Chronojump	Back squat (SM)	ICC	0.989	0.984			MPV: 0.986	Chronojump is a valid alternative, considering the particular error margins for each exercise and velocity result.
		Bench press (SM)	ICC	0.05 m/s	0.04 m/s			0.04 m/s	
			SEE	0.998	0.992			MPV: 0.997	
			ICC	0.04 m/s	0.05 m/s			0.06 m/s	
Pérez-Castilla et al. (2019)	Chronojump	Prone bench pull (SM)	ICC	0.997	0.993			MPV: 0.994	High device validity
			SEE	0.05 m/s	0.04 m/s			0.03 m/s	
		Bench press (SM)	ICC		0.72–0.90				
			r		0.991				
Fitronic									
Fernandes et al. (2018)	FitroDyne	Bench press (SM)	r	0.79–0.99	0.86–0.94				Valid device. Not interchangeable with GymAware
		Back squat (SM)	r	0.99–1.00	0.94–1.00				
		Bent-over row (SM)	r	0.91–0.98	0.91–0.98				
		Back squat (FW)	SEE	0.022 m/s	0.073 m/s				
Mitter et al. (2019)	FitroDyne	Bench press (FW)	SEE	0.019 m/s	0.100 m/s				FitroDyne valid for all exercises, but noticeable SEE in all mean values.
		Deadlift (FW)	SEE	0.018 m/s	0.084				
Tendo Sports Machines									
Garnacho-Castaño et al. (2015)	Tendo Weightlifting Analyser System	Back squat (SM)	ICC	0.963	0.985	PP: 0.853 MP: 0.966			Adequate concurrent validity
		Bench press (SM)	ICC	0.963	0.989	PP: 0.905 MP: 0.968			
		Both exercises	R ²	0.924	0.956	PP: 0.837 MP: 0.913			
		Back squat Jump squat	SEE	0.194 m/s 0.135 m/s	0.046 m/s 0.157 m/s				
Lorenzetti et al. (2017)	Tendo Weightlifting Analyser System							TTPV: 0.041 m/s TTPV: 0.064 m/s	Questions about whether a device is “valid” clearly depends on the research question, respectively, on the magnitude on the values to be quantified.

(Continued)

Table 4. (Continued).

Ref.	Device	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
, 2018	FitroDyne	Bench press (FW)	ICC		0.977				Valid device. Large degree of random error (overestimate with 80% 1RM loads and underestimate with 40% 1RM loads).
				R ²	0.985				
Squats & Science									
Gonzalez et al. (2019)	Open Barbell System	Back squat (FW)	r	0.58–0.95	0.25–0.35				More substantial differences in peak velocity as loads were increased. Valid device, but not interchangeable with Tendo.
		Front squat (FW)	r	0.49–0.86	–0.27–0.60				
		Both exercises	ICC	0.790–0.979	0.974–0.998				
			SEE	0.040–0.119 m/s	0.007–0.015 m/s				
Speed4Lifts									
Pérez-Castilla et al. (2019)	Speed4Lifts	Bench press (SM)	ICC		0.81–0.94				Speed4Lift is an appropriate device for measuring the velocity.
			r		0.994				
Linear Velocity Transducers									
Ergotech									
García-Ramos et al. (2016)	T-Force	Jump squat (SM)	r	0.827–0.909	0.866–0.894	PP: 0.901–0.944 MP: 0.928–0.959	PF: 0.937–0.987 MF: 0.963–0.989		Valid device Higher as the load to overcome increased, especially from 75% of 1RM. Not interchangeable with the force plate
Pérez-Castilla et al. (2017)	T-Force	RSJ	r	0.65–0.92		PP: 0.85–0.95	PF: 0.88–0.94		Valid device Not interchangeable with force plate. Questions about whether a device is “valid” clearly depend on the research question, respectively, on the magnitude of the values to be quantified. High device validity.
		NRSJ	r	0.71–0.87		PP: 0.80–0.91	PF: 0.90–0.95		
	T-Force	Back squat	SEE	0.151 m/s	0.070 m/s			TTPV: 0.026 m/s	
		Jump squat	SEE	0.263 m/s	0.167 m/s			TTPV: 0.045 m/s	
Pérez-Castilla et al. (2019)	T-Force	Bench press (SM)	ICC		0.77–0.95				
			r		0.995				

ICC: Intra-Class Correlation; FW: Free Weight; m/s: metres per second; MF: Mean Force; MP: Mean Power; NRSJ: Not restricted squat jump; PF: Peak Force; PP: Peak Power; r: Correlation coefficient; R2: Minimal products regression; RSJ: Restricted Squat Jump; SEE: Standard Error of the Estimate; SM: Smith machine; TTPV: Time To Peak Velocity; v: version; W: Watts.

Table 5. Reliability of the linear transducers for the measurement of the velocity of movement of the bar, power and force.

Ref.	Device (reliability type)	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
Linear Position Transducers									
Kinetic Performance Technology									
Askow et al. (2018)	GymAware (intra-device)	Back squat (FW)	ICC	0.982	0.966	PP: 0.993 MP: 0.972	PF: 0.979 MF: 0.992		Reliable device in all values of velocity, power and force. High reliability, due to low- moderate TEM for all variables.
Dorrell et al. (2019)	GymAware (intra-device)	Back squat	TEM	6.4–11.5 %	5.6–10.0 %		PF: 3.4–6.1 % MF: 0.5–0.9 %		
		Bench press	TEM	4.9–8.7 %	5.8–10.5 %		PF: 2.5–4.4 %		
		Deadlift	TEM	7.0–12.5 %	5.5–9.8 %		MF: 1.3–2.2 %		
Fernandes et al. (2018)	GymAware (intra-device)	Bench press (SM) Back squat (SM)	CV CV	1.1–6.1 % 2.0–4.8 %	1.2–6.3 % 2.1–11.2 %				GymAware is reliable for measuring peak and mean velocity. The data collected in squat with high loads (80% 1RM) should be interpreted with caution
Grgic et al. (2020)	GymAware (intra-device)	Bent-over row (SM) Squat	CV ICC	1.6–6.7 % 0.82–0.95	2.4–6.0 % 0.78–0.95	PP: 0.96–0.99 MP: 0.94–0.99			Reliable device Consistent reliability across the entire load spectrum (30–90% 1RM)
			CV	2.3–5.1 %	2.4–9.7 %	PP: 3.5–6.1 % MP: 2.8–6.9 %			
			SEM	0.04–0.08 m/s	0.03–0.06 m/s	PP: 94–176 W MP: 34–78 W			
		Deadlift	ICC	0.79–0.95	0.63–0.84	PP: 0.91–0.99 MP: 0.93–0.99			
			CV	2.2–8.3 %	3.3–10.6 %	PP: 3.0–8.8% MP: 2.8–9.3%			
			SEM	0.03–0.08 m/s	0.03–0.05 m/s	PP: 44–115 W MP: 20–56 W			

(Continued)

Table 5. (Continued).

Ref.	Device (reliability type)	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
Lorenzetti et al. (2017)	GymAware (intra-device)	Back squat	ICC	0.957	0.958			TTPV: 0.990	High device reliability, especially for monitoring through mean velocity and time to peak velocity.
		Jump squat	ICC	0.852	0.783			TTPV: 0.701	
Orange et al. (2020)	GymAware (intra-device)	Back squat (FW)	ICC	0.65–0.79	0.72–0.83	PP: 0.58–0.84 MP: 0.77–0.82			Great reliability in velocity measurement, especially for mean velocity 40–90% 1RM squat.
			SEM	0.06–0.09 m/s	0.03–0.05 m/s	PP: 196.4– 250.4 W MP: 73.0– 102.5 W			
		Bench press (FW)	ICC	0.68–0.82	0.64–0.87	PP: 0.64–0.91 MP: 0.81–0.91			Also especially reliable for peak velocity 40–80% 1RM.
			SEM	0.05–0.13 m/s	0.03–0.09 m/s	PP: 38.7– 78.0 W MP: 27.1– 52.8 W			
Thompson et al. (2020)	GymAware (intra-device)	Back squat (FW)	CV	4.9–22.0 %	2.9–13.6 %			High reliability, but CV and TEM increased as relative load increased (mostly at 90% and 100% 1RM)	
		Power clean	TEM CV	0.07–0.15 m/s 2.5–4.0 %	0.02–0.04 m/s 2.2–8.9 %				
			TEM	0.05–0.09 m/s	0.03–0.08 m/s				
Boscosystem									
Courtel-Ibáñez et al. (2019)	Chronojump (inter-device)	Back squat (SM)	ICC CV	0.996 1.8 %	0.991 3.6 %			MPV: 0.991 MPV: 3.9 %	Chronojump optimal alternative to T-Force.
		Bench press (SM)	SEM	0.03 m/s	0.03 m/s			MPV: 0.03 m/s	
			ICC	1.000	0.995			MPV: 0.995	
		Prone bench pull (SM)	CV	1.4 %	4.7 %			MPV: 5.2 %	
			SEM	0.02 m/s	0.04 m/s			MPV: 0.04 m/s	
			ICC	1.000	0.995			MPV: 0.995	
			CV	2.4 %	3.3 %			MPV: 3.4 %	
		SEM	0.04 m/s	0.04 m/s			MPV: 0.04 m/s		

(Continued)

Table 5. (Continued).

Ref.	Device (reliability type)	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
Pérez-Castilla et al. (2019)	Chronojump (intra-device)	Bench press (SM)	ICC	0.72–0.90					The data is not interchangeable with T-Force either or Speed4Lifts.
			CV	2.09–6.24 %					
Tendo Sports Machines									
Garnacho- Castaño et al. (2015)	Tendo Weightlifting Analyser System (intra-device)	Back squat (SM)	ICC	0.966	0.982	PP: 0.922 MP: 0.969 PP: 13.0 % MP: 10.9 %			High test-retest reliability
			CV	9.6 %	8.5 %	PP: 0.970 MP: 0.983 PP: 13.2 % MP: 13.7 %			
		Bench press (SM)	ICC	0.988	0.977				
			CV	9.0 %	9.6 %				
Lorenzetti et al. (2017)	Tendo Weightlifting Analyser System (intra-device)	Back squat	ICC	0.932	0.963		TTPV: 0.985		High reliability of the 3 devices, especially for monitoring through mean velocity and time to peak velocity.
		Jump squat	ICC	0.860	0.770		TTPV: 0.604		
Unimeasure									
Hansen, et al. (2011a)	HX VPA-200 (intra-device)	Jump squat	ICC			RFD: 0.75		TTPV: 0.90 TTPP: 0.94 TTPV: 9.4% TTPP: 10.7%	Low reliable in RFD.
			CV			RFD: 13.9 %			
Hansen et al. (2011b)	HX VPA-200 (intra-device)	Jump squat	ICC			RFD: 0.33	PF: 0.88	TTPF: 0.95	Reliable in peak force. Not reliable in RFD.
			CV			RFD: 44.5 %	PF: 4.8 %	TTPF: 14.3 %	
Speed4Lifts									
(Continued)									

(Continued)

Table 5. (Continued).

Ref.	Device (reliability type)	Exercise	Statistical test	Velocity variables		Power variables	Force variables	Acceleration variables	Lessons learned and concluding remarks
				Peak velocity	Mean velocity				
Ergatech									
Courel-Ibáñez et al. (2019)	T-Force (inter-device)	Back squat (SM)	ICC	0.999	0.999			MPV: 0.999	T-Force is very reliable to measure velocity.
			CV	0.8 %	1.0 %			MPV: 1.1 %	
		Bench press (SM)	SEM	0.01 m/s	0.01 m/s			MPV: 0.01 m/s	
			ICC	1.000	1.000			MPV: 1.000	
			CV	0.6 %	1.4 %			MPV: 1.3 %	
García-Ramos et al. (2016)	T-Force (intra- device)	Prone bench pull (SM)	SEM	0.01 m/s	0.01 m/s			MPV: 0.01 m/s	T-Force is very reliable to measure velocity.
			ICC	1.000	0.998			MPV: 0.998	
		Jump squat (SM)	CV	0.8 %	2.1 %			MPV: 1.9 %	
			SEM	0.01 m/s	0.02 m/s			MPV: 0.02 m/s	
			ICC	0.97	0.88	PP: 0.94 MP: 0.86 PP: 5.3 % MP: 10.1 %	PF: 0.97 MF: 0.97 PF: 3.3 % MF: 3.2 %		
Lorenzetti et al. (2017)	T-Force (intra-device)	Back squat	ICC	0.933	0.970			TTPV: 0.985	High device reliability, especially for monitoring through mean velocity and time to peak velocity.
		Jump squat	ICC	0.810	0.724			TTPV: 0.655	
Martínez-Cava et al. (2020)	T-Force (inter-device)	Back squat (SM)	ICC	1.000	1.000			MPV: 1.000	Very reliable T-Force across the entire velocity spectrum
			CV	0.46 %	0.44 %			MPV: 0.58 %	
Pérez-Castilla et al. (2019)	T-Force (intra-device)	Bench press (SM)	SEM	0.01 m/s	0.01 m/s			MPV: 0.01 m/s	T-Force was no more reliable than the Chronojump and Speed4lift.
			ICC	1.000	1.000			MPV: 1.000	
		Bench press (SM)	CV	0.45 %	0.55 %			MPV: 0.62 %	
			SEM	0.01 m/s	0.01 m/s			MPV: 0.01 m/s	
			ICC	0.77–0.95	1.82–4.90 %				

CV: % coefficient of variation; FW: Free weight; ICC: Intra-Class Correlation; m/s: metres per second; MF: Mean Force; MP: Mean Power; MPV: Mean Propulsive Velocity; PF: Peak Force; PP: Peak Power; RFD: Rate of Force Development; RM: Repetition Maximum; SEM: Standard Error of Measurement; SM: Smith machine; TEM: Typical Error of Measurement; TTPV: Time To Peak Velocity; TTPP: Time to peak power; TTPV: Time To Peak Velocity; W: Watts

high: 0.70–0.90, and almost perfect: > 0.90 (Hopkins et al., 2009)). Additionally, the least squares regression (R^2) (higher regression = greater accuracy) was considered to know the degree of correlation existing with the results of the ‘gold standard’. Using these statistical methods and their values as a standard, the results of the present systematic review can be unified.

Validity of linear position transducers (LPT)

The values extracted from LPT highlighted them as suitable technologies to assess motion during non-plyometric resistance exercises (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Lorenzetti et al., 2017; , 2018; Mitter et al., 2019; Pérez-Castilla et al., 2019; Thompson et al., 2020) (Askow et al., 2018; Fernandes et al., 2018; Lorenzetti et al., 2017; Pérez-Castilla et al., 2019; Thompson et al., 2020), independently of load spectrum (20–90% 1RM), or the type of exercise (back squat (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Lorenzetti et al., 2017; Mitter et al., 2019; Thompson et al., 2020), bench press (Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; , 2018; Mitter et al., 2019; Pérez-Castilla et al., 2019), prone bench pull (Courel-Ibáñez et al., 2019), power clean (Thompson et al., 2020), bent-over row (Fernandes et al., 2018) and deadlift (Dorrell et al., 2019; Mitter et al., 2019)). However, the critical factor for the accuracy of LPT could be the way in which the transition to the concentric contraction regime occurs.

Chronojump, GymAware, Tendo Weightlifting Analyser System, and FitroDyne (Tendo) were valid LPT during non-plyometric exercises with isometric pause of 0.5–1.5 seconds prior to each concentric muscle action ($ICC = 0.853–0.997$; $R^2 = 0.837–0.990$; $SEE = 0.014–0.100$) (Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Garnacho-Castaño et al., 2015; , 2018; Mitter et al., 2019) and without those isometric pauses ($ICC = 0.720–0.998$; $r = 0.79–1.00$; $R^2 = 0.86–0.99$; $SEE = 0.04–0.194$ m/s) (Askow et al., 2018; Fernandes et al., 2018; Lorenzetti et al., 2017; Pérez-Castilla et al., 2019; Thompson et al., 2020), highlighting a high consistency with proposed gold standard (Raptor 3D Motion Capture (Thompson et al., 2020); Rapture-E 3D (Dorrell et al., 2019); 3D Eagle Motion (, 2018), Vicon 3D (Lorenzetti et al., 2017) Qualysis Motion Capture System (Askow et al., 2018), TrioOptiTrack (Pérez-Castilla et al., 2019) (see Table 3). However, and despite LPT are valid for peak and mean measurement ($SEE = 0.04–0.100$ m/s), GymAware, FitroDyne (Fitronic), Tendo Weightlifting Analyser System, Chronojump, and Speed4Lift LPT devices extracted less accurate velocity values during those movement without isometric pause or eccentric controlled phase (0.04–0.194 m/s) (Askow et al., 2018; Lorenzetti et al., 2017) (see Table 4). The main reason that lead to this fact may be the greater sampling frequencies (the amount of data reported per second and expressed in Hertz (Hz) (Rico-González et al., 2020)) of the LTs when identify the real start of the push phase, which is dependent on the sampling frequency (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012, 2011). For example, Mitter et al. (2019) reported greater accuracy ($SEE = 0.018–0.022$ m/s) in the FitroDyne device (Fitronic; 100 Hz sample rate) compared to that reported by Courel-Ibáñez et al. (2019) ($SEE = 0.04–0.05$ m/s) using the Chronojump device (Boscosystem; 500 Hz sample rate), in very similar execution protocols (see Tables 3 and 4). In this scenario, the higher

the sampling frequency the greater the number of time points prior to the actual start will be integrated into the calculation runs, increasing measurement errors (Hori et al., 2007; Linthorne, 2001). This fact lead to the coaches to be careful with sampling frequencies. It would appear that lower LT sampling frequencies may provide greater validity than higher sampling frequencies when the assessment involves plyometric exercises, is performed in multiple planes and/or without isometric pauses prior to the concentric muscle action. However, no studies to date have investigated this hypothesis, which opens to highlight the most appropriate sampling frequency, as was occurred with other technologies (Rico-González et al., 2020).

In brief, LPT showed valid and accurate technologies to assess a movement velocity during non-plyometric exercises. However, sport scientist should be aware to the used sampling frequency and the sensitivity of these technologies measuring plyometric exercises or exercises without isometric pause, while future LPT developments could address this problem.

Validity of linear velocity transducers (LVT)

The reported results of the LVT announce an adequate validity for the measurement of velocity during plyometric (jump squat (García-Ramos et al., 2016; Pérez-Castilla et al., 2017)) and non-plyometric (back squat (Lorenzetti et al., 2017) and bench press (Pérez-Castilla et al., 2019)) exercises, independently to the load spectrum between 25% and 100% 1RM. However, different exercise's characteristic such as (i) isometric pause, (ii) plyometric, or (iii) non-plyometric nature were key factors in altering the validity values.

T-Force (Ergotech; 1000 Hz sample rate) was valid LVT in non-plyometric exercises without isometric pause or controlled eccentric phase (ICC = 0.77–0.95; SEE = 0.070–0.151 m/s) (Lorenzetti et al., 2017; Pérez-Castilla et al., 2017) in the velocity variables with respect to their gold standards (Vicon 3D) (Lorenzetti et al., 2017) and TrioOptitrack (Pérez-Castilla et al., 2019)) (see Tables 3 and 4). However, although T-Force remained valid, its precision values decreased during plyometric exercises ($r = 0.65$ – 0.92 ; SEE = 0.167–0.263 m/s) (García-Ramos et al., 2016; Lorenzetti et al., 2017; Pérez-Castilla et al., 2017) (see Table 3). As in previous section with LPT, the limitation to detect the real start of the thrust phase of the LT may be a key factor in measurement values. These measurement errors are maximised in plyometric exercises, and even more in the absence of an isometric pause before starting the concentric muscular action (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012, 2011). While the T-Force sample rate (1000 Hz) can be expected to increase measurement errors (Hori et al., 2007; Linthorne, 2001), direct measurement of velocity by LVTs (less manipulation of raw data) can help compensate for these errors (McMaster et al., 2014; Sánchez-Medina & González-Badillo, 2011). However, future research should analyse the validity of LVTs of different sampling frequencies to determine which is the most appropriate, as well as the impact of including a pause between contractions in order to minimise errors.

In summary, the LVT, and specifically T-Force, may be a valid device for monitoring the velocity of plyometric and non-plyometric exercises. However, it should be noted that precision errors increase considerably in plyometric exercises, and especially without an isometric pause prior to each concentric muscular action.

Reliability

The most considered statistical method has been CV, ICC, TEM, and SEM. The scientific literature suggests that optimal CV values should be below 5% for most sporting events and physical performance tests (Courel-Ibáñez et al., 2019; Martínez-Cava et al., 2020; Orange et al., 2020), while the scale of the ICC values have been previously presented in the previous section. In the field of VBT, TEM and SEM values of the maximum concentric velocity of 0.07–0.10 m/s can affect the 1RM values by 5% (González-Badillo & Sánchez-Medina, 2010; Martínez-Cava et al., 2019; Sánchez-Medina & González-Badillo, 2011; Sánchez-Medina et al., 2017). All of these values can help standardise the analysis of the results.

Reliability of LPTs

The LPTs proved to be reliable in the evaluation of movement in non-plyometric exercises (back squat (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Garnacho-Castaño et al., 2015; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Orange et al., 2020; Thompson et al., 2020), squat (Fernandes et al., 2018; Grgic et al., 2020), bench press (Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Martínez-Cava et al., 2020; Orange et al., 2020; Pérez-Castilla et al., 2019), deadlift (Dorrell et al., 2019; Grgic et al., 2020), prone bench pull (Courel-Ibáñez et al., 2019), biceps curl (Jennings et al., 2005), bent-over row (Fernandes et al., 2018), power clean (Thompson et al., 2020)), over the entire load range investigated (<90% of 1RM). However, the resulting variations were sensitive to the mode of execution of the movement, as well as to the 2D (Smith machine) or 3D plane (free weight) where the exercise was performed.

GymAware, FitroDyne (fitronic), Chronojump and Speed4Lift proved to be reliable devices during non-plyometric exercises, keeping a moderate-excellent intra and inter-device consistency (ICC = 0.64–1.00) (Askow et al., 2018; Courel-Ibáñez et al., 2019; Dorrell et al., 2019; Fernandes et al., 2018; Garnacho-Castaño et al., 2015; Grgic et al., 2020; Jennings et al., 2005; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Orange et al., 2020; Pérez-Castilla et al., 2019; Thompson et al., 2020) (see Table 5). However, the variations and measurement errors were, from smaller to larger, in protocols with isometric pause (CV = 1.4–4.7%; SEM = 0.02–0.04 m/s) (Courel-Ibáñez et al., 2019), followed by protocols with eccentric phase control (CV = 0.86–13.7%; SEM = 0.01–0.08 m/s; TEM = 4.9–10.5%) (Dorrell et al., 2019; Garnacho-Castaño et al., 2015; Grgic et al., 2020; Martínez-Cava et al., 2020), followed by protocols with absence of both phases (CV = 1.1–22%; SEM = 0.03–0.13 m/s; TEM = 0.02–0.15 m/s) (Askow et al., 2018; Fernandes et al., 2018; Lorenzetti et al., 2017; Orange et al., 2020; Pérez-Castilla et al., 2019; Thompson et al., 2020) (see Tables 3 and 5). As we mentioned previously, this could happen due to the limitation of the LTs to detect the real start of the concentric phase (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012, 2011), being also dependent on the sampling frequency (Hori et al., 2007; Linthorne, 2001). Furthermore, as an immediate consequence, the highest values of variation and measurement errors were recorded in the mean variables (CV = 1.00–11.2%) with respect to the peak (CV = 0.6–6.7%) (Courel-Ibáñez et al., 2019; Fernandes et al., 2018; Grgic et al., 2020; Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019; Thompson et al., 2020). Therefore, it is recommended to be guided by the peak values of the variables of interest, since they are less dependent on the addition of previous time points (Hori et al., 2007; Linthorne, 2001). LPTs

were also sensitive to displacement in multiple planes, also reported in previous studies (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012, 2011). For example, Thompson et al. (2020) and Orange et al. (2020) reported greater variations and measurement errors in peak values (CV = 4.9–22%; SEM = 0.06–0.09 m/s; TEM = 0.07–0.15 m/s) and mean values (CV = 2.9–13.6%; SEM = 0.03–0.05 m/s; TEM = 0.02–0.04 m/s) for performing back squat with free weight than the study by Fernandes et al. (2018) (Peak values: CV = 2.0–5.9%; Mean values: CV = 1.6–11.2%), for the same exercise with an execution protocol identical to the previous ones. This trend was generalised in the rest of the studies analysed in exercises of free weight (Dorrell et al., 2019; Grgic et al., 2020; Lorenzetti et al., 2017; Orange et al., 2020; Thompson et al., 2020) compared to exercises with the smith machine (see Table 5). According to previous studies, these variations may come from the alteration of joint moments and joint angular velocities, generated by the application of a load, and which triggers changes in the kinematics of the bar (Kellis et al., 2005; Kristiansen et al., 2019; Thompson et al., 2020). Pending a stronger body of evidence, plyometrics appear to maximise the errors described above (CV = 4.8–44.5%), even in protocols with isometric pause prior to concentric muscular actions (Hansen et al., 2011a, 2011b).

In summary, most LPT devices proved to be reliable for monitoring non-plyometric exercises. However, all of them were sensitive to the way the exercise was carried out and the plane in which it was developed.

Reliability of LVTs

LVTs proved to be reliable for the evaluation of movement in plyometric exercises (jump squat (Ferro et al., 2019; García-Ramos et al., 2016; Lorenzetti et al., 2017)) and non-plyometric exercises (back squat (Courel-Ibáñez et al., 2019; Lorenzetti et al., 2017; Martínez-Cava et al., 2020); bench press (Courel-Ibáñez et al., 2019; Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019) and prone bench pull (Courel-Ibáñez et al., 2019)), in a load spectrum of 25–100% 1RM (Courel-Ibáñez et al., 2019; Ferro et al., 2019; García-Ramos et al., 2016; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019). However, the reliability values were slightly sensitive to the plyometric or non-plyometric exercise nature.

T-Force and SmartCoach were reliable in plyometric (Ferro et al., 2019; García-Ramos et al., 2016; Lorenzetti et al., 2017) and non-plyometric exercises (Courel-Ibáñez et al., 2019; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019), both with isometric pause and in the absence of it (ICC = 0.72–1.00; CV = 0.44–4.90%; SEM = 0.01–0.04 m/s; TEM = 0.06 m/s) in all velocity variables, except in MV in jump squat with pause between contraction phases (CV = 8.3%) (García-Ramos et al., 2016) (see Tables 3 and 5). However, and although it remained reliable, the variations and measurement errors of T-Force were greater in plyometric exercises (ICC = 0.724–0.966; CV = 2.87–8.3%) (Ferro et al., 2019; García-Ramos et al., 2016; Lorenzetti et al., 2017) compared to non-plyometric exercises (ICC = 0.932–1.000; CV = 0.44–4.90%; SEM = 0.01–0.02 m/s) (Courel-Ibáñez et al., 2019; Lorenzetti et al., 2017; Martínez-Cava et al., 2020; Pérez-Castilla et al., 2019) (see Table 5). The fact that T-Force exhibits low measurement errors, even without isometric pause or in plyometric exercises, may lie in the use of a 14-bit resolution analogue-to-digital data acquisition board (Courel-Ibáñez et al., 2019; García-Ramos et al., 2016; Pérez-Castilla et al., 2019). In this way, the passage of the kinematic data through a depth of 14 bits increases the quality of the same (Filoux et al., 2011), eliminating noise from conflicting and sensitive

time points, such as the beginning or the end of each contraction regime (Cormie et al., 2007; Król & Gołaś, 2017; Swinton et al., 2012). If we add to this the subsequent passage through a low-pass Butterworth digital filter with a cut-off frequency of 10 Hz, T-Force offers us a considerable technological error minimisation (Courel-Ibáñez et al., 2019; Weakley et al., 2021). Given the scarcity of data in this regard, future research should evaluate the impact caused by the absence of pause between contraction regimes, as well as the performance of exercises with free weight, on the reliability of the LVT.

Definitely, the LVTs seemed to be reliable devices for the evaluation of plyometric and non-plyometric exercises with isometric phase between contraction regimes made in Smith machine, achieving higher reliability values in the latter.

In conclusion, strength and conditioning coaches should be aware that while LPTs and LVTs seem to be valid and reliable devices for VBT monitoring, there are technological factors that can increase measurement errors. Among these factors are the lower accuracy in explosive movements, the difficulty of detecting the actual beginning of the concentric phase of the movement, as well as the sensitivity shown in the calculation results when it comes to horizontal displacements. Therefore, when using this type of force monitoring device, the professionals involved must take into account the risks involved in performing a certain type of exercise, the plane in which it is performed, as well as the way in which it is performed the movement. Future developments of these technologies should try to solve these measurement errors. Due to data processing systems and direct calculation of velocity, it appears that LVTs seem to have lower measurement errors in all parameters relative to LPT under similar conditions. Finally, it is recommended to review other aspects that were not studied in depth in this review, such as the brand and version of each device.

Conclusion

LPTs may be valid and reliable devices for the assessment of velocity of movement during non-plyometric exercises, regardless of the load range. However, these devices lose some validity and reliability in protocols without a pause prior to the concentric contraction regime, with greater measurement errors the higher the sampling frequency. Furthermore, the reliability of the LPTs was sensitive to the performance of exercises in multiple planes (free weight). On the other hand, the LVTs were valid and reliable to monitor the velocity of movement in plyometric and non-plyometric exercises, although lower precision and reliability values were reported in the former. Furthermore, despite showing greater variations and measurement errors in the absence of isometric pause, as was the case with LPTs, LVTs managed to minimise technological errors, due to factors inherent to their technology. In all cases, the peak values presented more reliable and precise values than the mean values, most affected by the technological errors of the LT. Future research should analyse the behaviour of the LPTs in the face of plyometric exercises, developed in different planes and with different execution protocols. The optimal frequency of LPT and LVT should also be investigated, taking into account the characteristics of the assessment protocol.

Disclosure statement

The authors declare that they have no competing financial interests.

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